

Surface Defect Detection Based on Machine Learning and Data Augmentation

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Abstract—The development of industrial automation and intelligent manufacturing has promoted the research of computer vision and machine learning in the automated detection of surface defects in metal and plastic products, to solve the problems of low efficiency and high inconsistency in traditional manual detection. In response to such demands, this paper proposes a lightweight model that can be used for high-precision industrial defect detection. Data Preprocessing First, we use Python to write programs to read images. Secondly, image features are extracted. In this paper, the HOG algorithm is used to extract the features of the image, and the principal component analysis (PCA) is used to reduce the dimension of the feature. Finally, we trained a defect target detection model based on grid division and LightGBM, which we call the Grid-HOG-PCA-LightGBM model. The recognition accuracy of the model can reach 98%, and it only takes up 4000B memory size. Our models not only enable automated inspection, increasing inspection efficiency and accuracy; they also reduce costs in the production process by enabling continuous and efficient inspection.

Keywords—Defect detection, Light Gradient Boosting Machine, Histogram of Oriented Gradient, Principal Component Analysis

I. INTRODUCTION

In recent years, with the booming development of industrial automation and intelligent manufacturing, product quality monitoring and assurance has become an important concern for various industries. Especially for metal and plastic products, surface defects not only affect the visual aesthetics of the product but also pose a potential threat to its functional performance and durability. Traditional manual inspection methods are time-consuming and inefficient, and due to the influence of human subjective factors, which can easily lead to inconsistency in the inspection results. To solve these problems, automated surface defect detection based on computer vision and machine learning has gradually become a hot field in academic research and industrial practice.

Antecedent scholars have adeptly leveraged deep learning methodologies to construct a model expressly tailored for surface defect detection, asserting its discriminative efficacy even in instances characterized by scant training data^[1]. The existing research results indicate that the advanced TCM-YOLO algorithm achieved an average accuracy of 81.8% on the strip defect dataset of Northeastern University, which is

7.4 percentage points higher than the traditional YOLO algorithm. The accuracy of this algorithm reached 78.3%, which is 8.9 percentage points higher than the original algorithm. In addition, the detection speed of TCM-YOLO algorithm reached 61.73 frames per second (FPS)^[2]. However, the TCM-YOLO algorithm cannot maintain robust performance when facing defect types that were not previously considered.

In response to the limitations of previous research fields, the Grid HOG PCA LightGBM model is based on the HOG algorithm and the LightGBM model, proposing feature extraction based on the HOG algorithm and reducing feature dimensionality through dominant analysis. This paper also introduces two innovative cores based on the LightGBM model: firstly, each weak model in GBDT focuses on correcting the errors of the previous model, thereby forming a strong model. Secondly, CatBoost introduces common category feature processing methods such as single hot encoding, as well as an innovative technique based on learning rate sorting to avoid the mutual influence between category features and obtain more accurate prediction results. In light of these considerations, our endeavor is grounded in the development of a user-friendly mathematical model tailored for handheld devices, characterized by judicious computational and storage demands, and distinguished by its adeptness in navigating uncharted territory with respect to defect typologies.

II. DATA PREPROCESSING AND PREPARATION

The data in this paper is sourced from <http://mcm.tzmcm.cn/>

A. Data Reading

The image data in the attachment is given in the form of a folder, and the format of the image is a .jpg file. Use Python's OS module and OpenCV module to write programs and read data. Finally, 399 images in the attachment are stored in an array format^[3].

The images involved in this modeling are three-channel digital images, which are represented by the RGB color space model. The label of the picture is a gray image, which is a two-dimensional array composed of 0 and 1.

B. Image Resize

The size of the image in the attachment is not uniform. According to the rough estimation, the height-width ratio of the picture is about 2 : 1, so the OpenCV module is used to convert the size of the picture to 1000x500. The data processing results are shown in Figure 1.

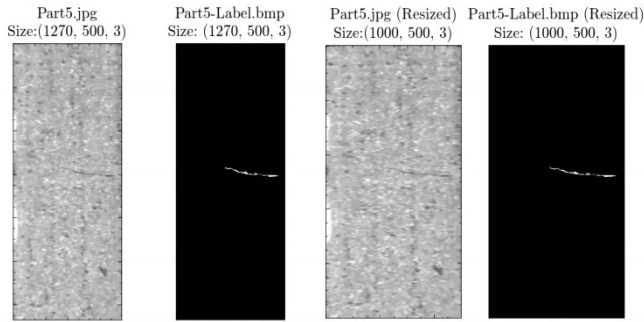


Figure 1: Image resize

C. Image Grayscale

The color digital image has three channels R, G, and B. To facilitate data analysis, the image is usually converted into a grayscale image. The grayscale image refers to a two-dimensional array, and each pixel of the two-dimensional array condenses the R, G, and B pixels under the RGB color space model. Specifically, the formula of image graying is:

$$\text{Image}_{\text{Grey}} = 0.30 \times \text{Image}_{\text{Red}} + 0.59 \times \text{Image}_{\text{Green}} + 0.11 \times \text{Image}_{\text{Blue}} \quad (1)$$

Since the sum of the weights of different colors in the R, G, and B color space model is 1.0, the pixel range of the grayscale image is still [0,255]. In addition, each pixel of the grayscale image needs to be converted into an integer. The data processing results are shown in Figure 2.

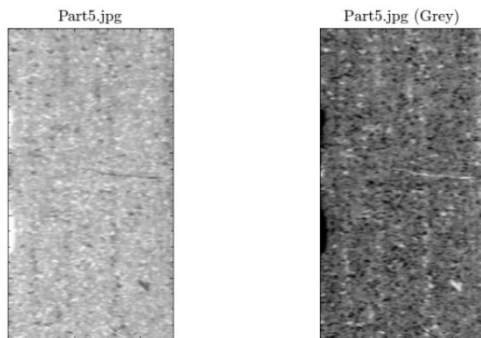


Figure 2: Image grayscale

D. Image Augmentation Based on Noise

A total of 399 images are provided to us in the attachment, and the number is small. To improve the quality of the data set, the image data set should be augmented. Due to the industrial fault detection data set of the data set involved in this paper, the images involved will have more noise, so it can be considered to add different intensity noise to the image, to achieve image augmentation.

1) Gaussian Noise

Specifically, first, divide each pixel of the image by 255, then add different intensities of normal distribution noise, and then multiply each pixel by 255. The data processing results

are shown in Figure 3.

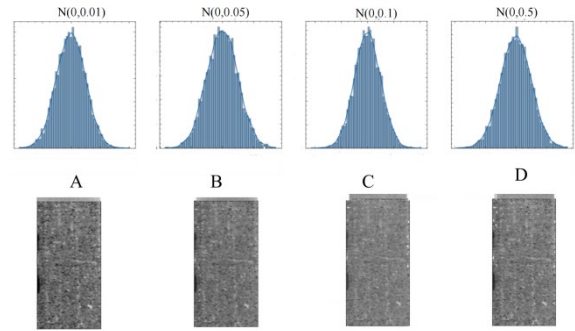


Figure 3: Images with Gaussian noise. (A) With noise N(0,0.01); (B) With noise N(0,0.05); (C) With noise N(0,0.1); (D) With noise N(0,0.5)

2) Laplace Noise

The Laplace distribution is a more acute bell-shaped distribution than the Gaussian distribution, and the Laplace distribution is more common in industrial data sets.

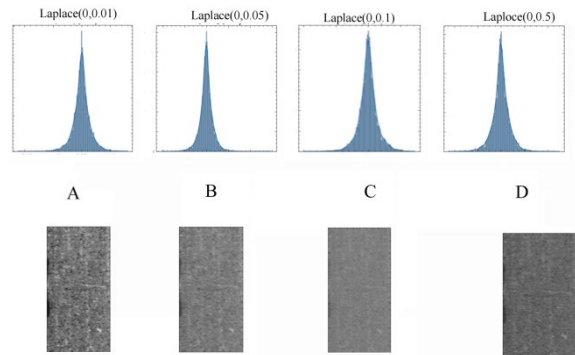


Figure 4: Images with Laplace noise. (A) With noise Laplace(0,0.01); (B) With noise Laplace(0,0.05); (C) With noise Laplace(0,0.1); (D) With noise Laplace(0,0.5)

Finally, this study added Gaussian noise and Laplace noise twice for each image and expanded the dataset 4 times. The data processing results are shown in Figure 4.

III. IMAGE FEATURE EXTRACTION AND DIMENSIONALITY REDUCTION

A. Feature Engineering

1) Feature extraction based on the HOG algorithm

Histogram of Oriented Gradients (HOG) is a feature extraction algorithm widely used in computer vision, especially for object detection tasks. Developed by Navneet Dalal and Bill Triggs in 2005, HOG has been instrumental in various applications, including pedestrian detection and face recognition. The HOG algorithm operates by dividing an image into smaller portions, called cells. Each of these cells then undergoes a process to compute the gradient or direction of intensity change, which is subsequently represented in a histogram format. This histogram provides a quantitative representation of the distribution of directed gradients within the cell^[4].

After computing the gradient histogram for each cell, these histograms are then combined to form a larger feature descriptor, known as a block. This process of combining histograms from multiple cells into blocks helps to normalize the features and provide robustness against local variations in illumination, shadowing, and edge contrast.

The final step involves concatenating the descriptors from all blocks to form the HOG feature vector, which serves as a robust representation of the entire image or the region of interest. This vector encapsulates structural and appearance information vital for object recognition tasks. The data processing results are shown in Figure 5.

In this paper, the HOG feature extraction algorithm is implemented using Python's open-source image processing package *image*, and the HOG feature is visualized in the form of a feature map. The results are as follows.

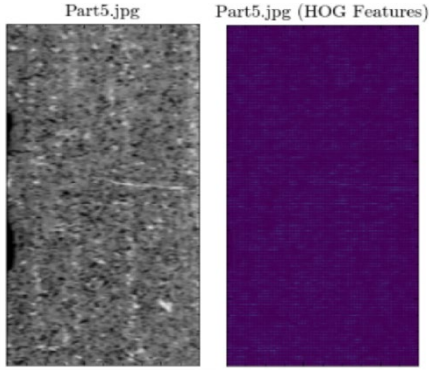


Figure 5: Feature extraction based on HOG algorithm

2) Feature dimension reduction based on PCA

The features extracted by the HOG algorithm have a higher dimension, which will bring a greater computational burden to the model. Therefore, it is necessary to use data dimension reduction algorithm to reduce the dimension of features.

This paper uses principal component analysis (PCA). Principal Component Analysis (PCA) is a widely used statistical technique that aims to simplify high-dimensional datasets by identifying patterns and correlations within the data. It does this by transforming the original variables into a smaller set of principal components, which represent the most important features of the dataset.

PCA works by first identifying the direction of maximum variance in the data and projecting the data onto a new axis. This first principal component captures the greatest amount of information, or variance, in the data. The second principal component is then chosen to be orthogonal to the first and captures the remaining variance not explained by the first component. This process is repeated, creating a new axis each time that captures the maximum remaining variance, until a desired number of principal components are obtained.

These principal components are ordered by the amount of variance they explain, with the first component explaining the most and each subsequent component explaining less. By selecting only the top principal components, which often account for a large proportion of the total variance, it is possible to greatly reduce the dimensionality of the dataset while retaining most of the important information.

The principal component analysis algorithm is used to reduce the high-dimensional feature vector to the low dimension. In this paper, the dimension length of the target feature dimension is 30. When the feature dimension is reduced to 30, the variance interpretation rate and cumulative variance interpretation rate of the model are shown in the following figure.

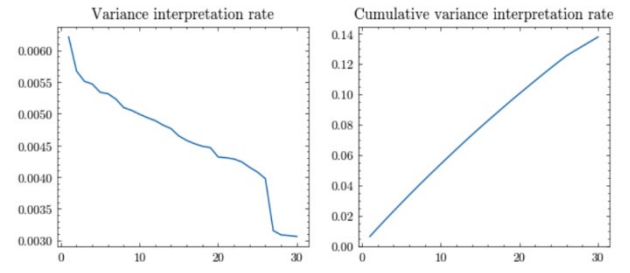


Figure 6: Feature extraction based on HOG algorithm

The data processing results are shown in Figure 6. It can be seen that although the feature dimension is greatly reduced, the feature vector obtained after dimensionality reduction can also reflect the characteristics of the original image to a large extent.

B. Models

Decision Trees have the advantages of intuitiveness and ease of explanation, but they are prone to overfitting and are sensitive to noise. Random Forest improves stability and accuracy by integrating multiple decision charts, reducing the risk of overfitting, but the computation speed is slower^[5]. Extra Trees further accelerate the training process and improve efficiency by randomly selecting splitting points, but sometimes may not be as accurate as other methods. GBDT (Gradient Boosting Decision Tree) significantly improves prediction accuracy by gradually optimizing the model, but may overfit and incur high computational costs^[6]. LightGBM significantly improves training speed and large-scale data processing capabilities through efficient leaf node optimization and parallel processing based on GBDT, but may not perform as well as GBDT on small datasets^[7]. CatBoost is particularly adept at handling category features and has improved prediction accuracy through innovative category feature processing techniques, but it heavily relies on data preprocessing. Overall, these algorithms have their own advantages and disadvantages under different data characteristics and computational requirements. Choosing the appropriate algorithm can better solve practical problems.

Overall, these algorithms rely on decision trees in the process of building models, but through different strategies and innovative points, they provide flexible and diverse solutions to address data features and computational challenges in different scenarios.

C. Model Accuracy Evaluation

In this paper, the following metrics are used to evaluate model performance:

Accuracy Proportion of correctly predicted samples to the total number of samples.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (2)$$

where TP denotes the number of true cases, i.e., the number of positive samples correctly predicted as positive; T N denotes the number of true counterexamples, i.e., the number of negative samples correctly predicted as counterexamples; FP denotes the number of false positive cases, i.e., the number of negative samples incorrectly predicted as positive; and F N denotes the number of false counterexamples, i.e., the number of positive samples incorrectly predicted as counterexamples.

Precision: The number of samples correctly predicted as positive cases as a proportion of the number of all samples predicted as positive cases. Precision metrics in binary classification reflect the accuracy of model predictions. The precision metric is a type of categorization accuracy that measures the proportion of positive examples (Positive) that the model correctly predicts. In a binary classification problem, if the accuracy of the model is 100%, it means that the model predicts all Positive examples exactly correctly. If the accuracy is 50%, it means that the model correctly predicts half of the Positive cases. The precision metric is very important in determining how well the model works in dealing with real-world problems, especially in cases where high precision classification is required.

$$\text{Precision} = \frac{TP}{TP+FP} \quad (3)$$

Recall: The number of samples correctly predicted to be positive cases as a proportion of the number of samples of all positive cases. Recall (Recall) is another important evaluation metric in binary classification problems. It represents the proportion of positive examples correctly identified by the model. If the recall is 100%, it means that the model has successfully found all the positive examples. If the recall is 50%, it means that the model only found half of the positive examples.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (4)$$

F1 Score: The reconciled mean of precision and recall. The F1 metric is a statistical measure of the precision of a dichotomous classification model that takes into account both the accuracy and recall of the classification model. The F1 score can be thought of as a kind of weighted average of the model's accuracy and recall. Its maximum value is 1, and its minimum value is 0. The larger the value, the more likely it is that the model will find half the positive examples. The maximum value is 1 and the minimum value is 0. A larger value means a better model.

$$F1 = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

Training Time: The model training time refers to the time used from data input to the model, to the completion of model parameter fitting. In this paper, we do not take the data preprocessing time into the model training time. For traditional machine learning models, the training time is the HOG feature extraction time, PCA data dimensionality reduction time and ML model training time combined.

This paper calculates the training time using Python's time module, and the unit of training time is seconds (s).

Model Scale: The size of the storage space occupied by the model in the computer memory. Since the model in this paper is based on Python training, the size of the space occupied by Python objects is used to represent the size of the memory occupied by the model.

TABLE I MODEL EVALUATION

Model Name	Accuracy	Precision	Recall	F1-Score	Training Time	Scale
Decision Tree	0.8300	0.8864	0.9176	0.9017	0.0158s	1927B
Random Forest	0.8800	0.8763	1.0000	0.9341	0.3626s	67883B
Extra Trees	0.8800	0.8763	1.0000	0.9341	0.2641s	67849B
GBDT	0.8800	0.8763	1.0000	0.9341	0.2802s	10684B
LightGBM	0.8800	0.8763	1.0000	0.9341	0.1095s	8373B
CatBoost	0.8800	0.8763	1.0000	0.9341	3.7200s	2622B

The data processing results are shown in Table I. In terms of the accuracy of model recognition, the Decision Tree model has the lowest accuracy and is not considered. In terms of the size of the space occupied by the model, the CatBoost model and the LightGBM model have less space overhead. However, the CatBoost model takes too long to compute, so the LightGBM model is the best choice for this task.

IV. OPTIMIZATION OF COMPUTATIONAL COSTS IN TARGET DETECTION

A. Grid Model

In order to perform faster and more accurate defect location target detection, this paper first proposes the Grid Model. Grid Model means that the image is divided into several grids, and each grid is marked separately. If the grid contains defects, it is marked as 1, otherwise it is marked as 0. The figure 7 shows the principle of the Grid Model.

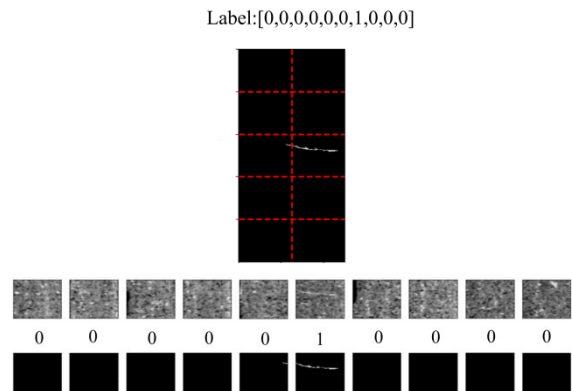


Figure 7: Grid model

The data processing results are shown in Figure 7. It can be seen that the Grid Model transforms the target detection task into an image recognition task, so that the model in Problem I can be used to solve the problem.

B. Grid LightGBM Model

We combine the Grid Model with the LightGBM model

established above, and call the model Grid LightGBM Model^[8]. The steps of training the model are described as follows^[9]:

Step I: Create a Grid Model. Each image is split into 10 100x100 small sub-images and marked separately.

Step II: Extract Features with HOG. The HOG algorithm is used to extract the feature vector of each sub-image.

Step III: Data dimensionality reduction with PCA. The PCA model is used to reduce the dimension of the feature vector extracted by the HOG algorithm to 30 dimensions^[10].

Step IV: Train LightGBM.

When the model works, the image is first converted into 10 sub-images, and each sub-image is identified by image recognition. If a defect is identified, the subgraph is marked. The following image shows the marking effect of the Grid LightGBM Model. The data processing results are shown in Figure 8.

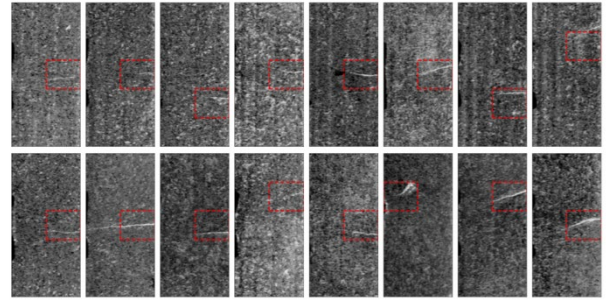


Figure 8: Marking effect of the Grid LightGBM Model

C. Model Accuracy Evaluation

The data processing results are shown in Table II. It can be seen that due to the reduction of sub-images, the accuracy of the LightGBM model is significantly improved, and the operation speed of the model and the space occupied by the model are significantly reduced.

TABLE II MODEL EVALUATION OF GRID LIGHTGBM MODEL

Model Name	Accuracy	Precision	Recall	F1-Score	Training Time	Scale
LightGBM	0.9800	0.9763	1.0000	0.9641	0.050s	4273B

To verify the generalization performance of the model, this paper tests the sensitivity of the model based on normal distribution random numbers^[11]. Specifically, normal distribution noise with different variances is added to the training set, and the performance of the model on the test set is tested to test the sensitivity of the model to noise. The data processing results are shown in Figure 9.

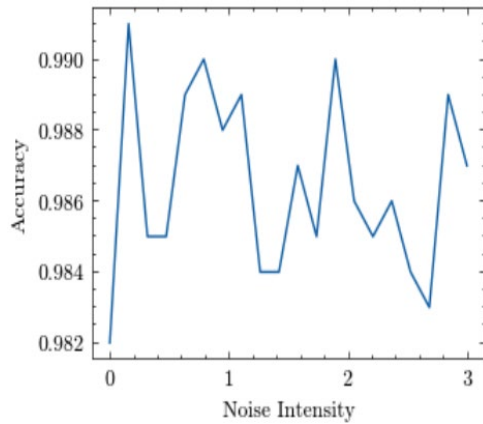


Figure 9: Sensitivity test

It can be seen that with the increase of noise intensity, the accuracy of the model on the test set fluctuates within a small range, indicating that the model has good generalization performance and anti-interference ability.

The biggest innovation of this paper is that this paper does not use the deep learning model for image recognition tasks and target detection tasks, but uses the most traditional machine learning model. In contrast, the model in this paper is far better than the deep learning model in terms of time and memory space. In order to further improve the operation speed of the model, the model can be optimized at the hardware level. For example, embedded programming techniques can be used to write faster HOG feature extraction algorithms.

V. CONCLUSIONS

This paper proposes a lightweight model that can be used for high-precision industrial defect detection. First, we encode each image into a feature vector with a length of 30. Then, we trained Random Forest, ExtraTrees, GBDT, LightGBM, CatBoost, and other integrated learning models, and used accuracy, precision, recall, F1 score, model running time, model memory size, and other indicators to evaluate the model. Finally, we find that LightGBM has high accuracy, and its operation time can reach the millisecond level. The memory size of the model is about 8000B, which is most suitable for deployment on mobile devices.

We trained a defect target detection model based on grid division and LightGBM, which we call the Grid-HOG-PCA-LightGBM model. The recognition accuracy of the model can reach 98%, and it only takes up 4000B memory size. Finally, we use the normal distribution perturbation to analyze the sensitivity of the model, which proves that the model has strong generalization ability. This proves the feasibility of our algorithm in the field of industrial image recognition.

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